

Data types and Variables

Identifiers

The names given to variables, functions, modules are called identifiers. These names can contain one or more characters such as upper case letters A-Z, lower case letters a-z, numbers 1-9, or symbols.

NOTE: Identifiers should not start with a number but the number can be part of the identifier.

Keywords

Words that have an already pre-defined meaning in python language are referred to as keywords. Because they already have a predefined meaning, they cannot be used by programmers as identifiers. An attempt to use a keyword as an identifier will result in an error.

The code below displays the keyword in python.

To get more information about the keywords, type `help(keyword)` e.g `help(False)` will give you information about the help keyword.

```
import keyword
list = keyword.kwlist
print(list)
```

Variables

A named memory place holder for holding different types of data for future manipulation is called a variable. To create a variable in python, you just give it a name and assign a value.

```
#For example;
number=10
#creates a variable named number and stores the value of 10 in it.
#Python does not require one to explicitly say the data type of the
variable when creating it.
```

Variable naming

The following are the naming conventions for variables 1. It should start with a letter 2. It can contain a number, a letter and a symbol 3. You should not use the keywords as variable names 4. The names of variable names are case sensitive, Number and number are different types of variables 5. Use lowercase letters for python variable names 6. Does not contain spaces. If you want to create a two word variable, you can separate them using an underscore. 7. Avoid using an underscore as the first letter of the variable 8. Make sure the variable names are descriptive

Variable assignments,

The = sign is used to assign a value to a variable

Variable_name=expression

```
#Example 2:
names="Malik"
```

Data types

Strings

A sequence of more than one character is called a string. A string can contain letters, numbers and symbols. In python strings can be represented using both single and double quotes. Several lines of strings can be represented using a triple quote.

```
#Example 1
string1="This is a string created using a double quote"
string1
```

```
#Example 2
string2='This is a string created using a single quote'
string2
```

```
#Example 3:
string3="""The reader is reading a string with several lines.
This is created using tripple quotes, this spans wide and is created
using a triple quote"""
string3
```

Empty string is created using empty quotes

```
#Example 4
empty=" "
empty
```

Basic string operations

Concatenating strings

To concatenate two strings, which means combining two strings, python uses the + operator

```
#Example 1:
first_name= "Eliud "
second_name="Mathews"
first_name+second_name
```

Str() Function

The string functions converts an object to a string

```
#Example 1
str(10) #Converts the number 10 to a string 10
```

String repetition

The * operator is used to repeat a string

```
#Example
first_name *3 #Prints the value stored in the variable first_name three
times
```

Built in functions used with strings

Python has the following built in functions for culculating the no of characters in a string, returning the max and the minimun character in the string len() : Culculates the legth of the string

max() :returns a character having highest ASCII value.

min() :returns a character having the lowest ASCII value

str.capitalize(): Capitalizes the strings first letter

str.lower(): makes string lower case

str.upper(): makes string upper case

```
len(second_name)# Returns the no of characters in the name 'ELiud' which is 7
```

```
max(second_name)
```

```
min(second_name)
```

```
second_name.capitalize()
```

```
second_name.upper()
```

```
second_name.lower()
```

Numbers

Integers and floating point are all categorized as numbers in python. An integer can be as long as the computer memory will allow. The accuracy for the floating point is 15

2 is an integer 2.5 is a floating point

```
number=7 # This is an integer
number
```

```
number=8
type(number)
```

```
number1=7.6 #This is a float
```

```
type(number1)
```

```
Activity 2: Reading activity
```

Boolean

Booleans are the variables that are true or false. The boolean values True and False are in the list of python keywords

None

None in python is usually used when there is no value.

Reading Input

All programs process data. This data can come from a file or from keyboard (input by the user of the program). In order to get the input from the user in python, the input() function is used. This is a predefined function, meaning you do not need to create the logic of how the function is working. You will learn about creating your own functions in the future modules

```
name=input()
```

In the above line, we have created a variable called data, and we are storing the value we are getting from the input() function . The input function returns a string as the input. In order to get numbers we can use casting to convert a string to an int or float. Python has predefined methods for converting data between the three different data types.

```
#Example
number_int=int(input()) #this will result in a value that is an integer.
number_int
```

```
number_float=float(input("Enter floating point"))
```

```
#Example : String format
```

```
names="Caren Lete"
```

```
other_name="Oumalama"
```

```
print(f' my names are {names} and my other name is {other_name}')
```